KANAN SHAH

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SUMMARY

Software Engineer with hands-on experience in AI, computer vision, and full-stack development, adept at designing scalable, high-performance systems. Proven ability to deliver innovative software solutions in fast-paced environments, eager to contribute to cloud-based physical security systems.

EDUCATION

California State University, Sacramento **Jan 2021 – May 2025**
Bachelor of Science in Computer Science

Coursework: Database Management Systems, Data Structures and Algorithms, Artificial Intelligence, Data Analytics and Mining Operating Systems, Machine Learning, Software Engineering, Object Oriented Programming

TECHNICAL SKILLS

Programming Languages:	Java, Python, JavaScript, SQL, Bash, PowerShell
Web Technologies & Databases:	HTML5, CSS, JavaScript, REST API, SQL, PostgreSQL, MySQL, MongoDB, Node.js, Django
Cloud & DevOps:	Azure, Docker, Kubernetes, Git, Maven
Frameworks:	Spring Boot, React.js, Next.js, Flask
Testing & Tools:	JUnit, Jest, Selenium, Postman, Jira, Trello, Microsoft Teams
Operating Systems:	Linux/UNIX, Windows Server
Security:	OAuth, SSL/TLS
Methodologies:	Agile, Software Development Lifecycle (SDLC)

EXPERIENCE

Software Engineering Intern | California Department Of Public Health, Sacramento **Feb 2024 – Dec 2024**

- Spearheaded front-end development of the MyInspection Buddy iOS app using Expo Go, integrating a Python Flask backend and diverse FDA/internal data sources to streamline environmental inspection workflows.
- Designed and implemented a novel panorama-to-cube-map algorithm using OpenCV, enabling stereoscopic 3D image capture and improving object detection accuracy by 20% in real-world field inspections.
- Embedded stereoscopic images into interactive web environments using Panellum, delivering immersive, plugin-free 3D training simulations accessible directly via browser links.
- Captured 2D projections from stereoscopic images to generate labeled training data for the YOLO object detection model, enhancing automated inspection capabilities.
- Automated medical device recall data pipelines using Python, ensuring state data integrity practices and optimizing environmental reporting efficiency

Software Engineer Fellow | HeadStarter AI , San Francisco **July 2024 – Sept 2024**

- Developed scalable web applications using React.js and Next.js, implementing features for intelligent Q&A and automated document generation, adhering to MVC architecture principles.
- Integrated REST and GraphQL APIs to synchronize legacy enterprise data with modern web applications, significantly enhancing system interoperability and supporting robust data flow for complex IT systems.
- Contributed to the full software development lifecycle, including analysis, design, and testing, ensuring alignment with user needs and system requirements.

PROJECTS

LuxPartyRents - Commercial Equipment Rental Website | Spring Boot, MySQL, JavaScript, Bootstrap, HTML/CSS

- Led development of a full-stack web application using JavaScript and Spring Boot, enabling real-time order processing and inventory management for a real-world client.
- Collaborated with stakeholders to design intuitive interfaces, ensuring scalability and adherence to client requirements through Agile methodologies.

Expense Receipt Scanner | JavaScript, React.js, Tesseract.js, Ollama LLM (2025)

- Developed a free AI-powered web application for fast, private expense tracking, enabling instant categorization of uploaded receipts using client-side OCR (Tesseract.js) and on-device LLMs (Ollama).
- Engineered a fully responsive and accessible UI with React.js, delivering interactive analytics dashboards and ensuring data privacy, while overseeing state management, category rules engine, and end-to-end unit testing.

Next AI-Support.io | Next.js, Pinecone, OpenRouter API, Retrieval-Augmented Generation (RAG)

- Developed an AI-powered chatbot leveraging Next.js, Pinecone, and OpenRouter API with Retrieval-Augmented Generation (RAG), providing contextual support and automated document generation.
- Ensured backend services aligned with user needs through structured testing and iterative feedback.

Car Insurance Predictor | Python, Flask, scikit-learn, Streamlit

- Developed a machine learning model using Python and scikit-learn to predict car insurance premiums, exposing the model through a Flask API and a Streamlit UI for user interaction.
- Ensured backend services aligned with user needs through structured testing and iterative feedback.